

 Marwadi University Marwadi Chandarana Group 	Marwadi University Faculty of Engineering & Technology Department of Information and Communication Technology	
Subject: DIGITAL VLSI TESTING (01CT0629)	Aim: Detection of both SA0 and SA1 faults on a VLSI circuit that represents 3–input ODD function.	
Experiment No: 1	Date: 04/03/2026	Enrollment No:92301733056

Aim:

1. To understand stuck-at-0 and stuck-at-1 fault using 3-input ODD function.
2. Implementation and visualization of stuck-at-0 and stuck-at-1 fault with the help of XOR gate simulator.

Simulation Platform: Virtual Lab

<https://vlsi-nitk.vlabs.ac.in/exp/3-input-odd-function/>

Theory:

Exclusive-OR (XOR)

The XOR gate, also known as Exclusive OR (sometimes written as EOR or EXOR), is constructed using AND-OR-INVERT (AOI) logic. It produces a HIGH (1) output when the number of HIGH inputs is odd. This means the output is true only if exactly one of the inputs is true, and false if both inputs are the same (either both 0 or both 1). XOR essentially checks for inequality between inputs. A simple way to remember XOR is: "one or the other, but not both."

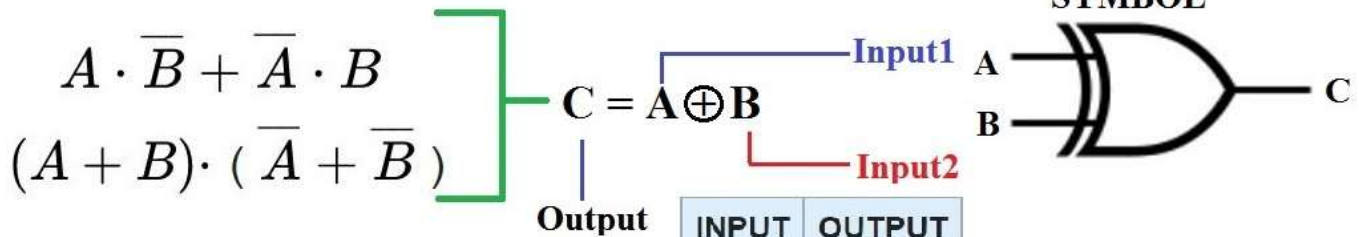
For three inputs, the XOR gate outputs 1 when either exactly one input is 1 or when all three inputs are 1. In general, for three or more inputs, the output is 1 if an odd number of the inputs are 1.



Schematic symbols for XOR gates

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BOOLEAN EXPRESSION



INPUT		OUTPUT
A	B	A XOR B
0	0	0
0	1	1
1	0	1
1	1	0

XOR

Faults in logical circuits

A fault is a physical defect in a circuit or system that may lead to malfunction. It is typically represented at the logic level to simplify analysis, serving as an abstract model of how a physical defect alters the circuit's logical behavior.

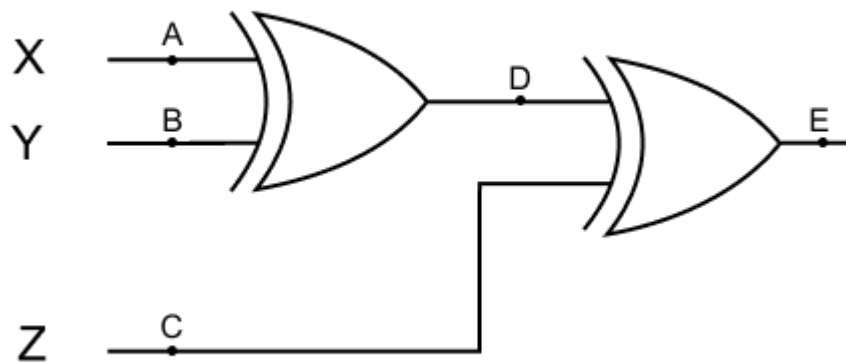
To streamline error detection and reduce complexity, logical errors are used to represent physical faults. The most widely used model is the **single stuck-at fault**, where a fault causes a signal (input or output) to be permanently fixed at logic 0 (stuck-at-0 or s-a-0) or logic 1 (stuck-at-1 or s-a-1).

Stuck-at fault (SAF)

This model simulates manufacturing defects by assuming certain signals or pins are permanently stuck at a logical '1', '0', or 'X'. For instance, an input may be assumed stuck at logic 1 during test generation

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to ensure a test pattern can detect a defect with that behavior. Similarly, it could be stuck at logic 0 to simulate a fault that prevents proper signal switching.



A, B, C, D, E are different points in the circuit. They can be stuck at 0 or 1 due to the fault.

$$\text{Output} = X \oplus Y \oplus Z$$

Truth Table

Truth Table for 3-Input Odd Function

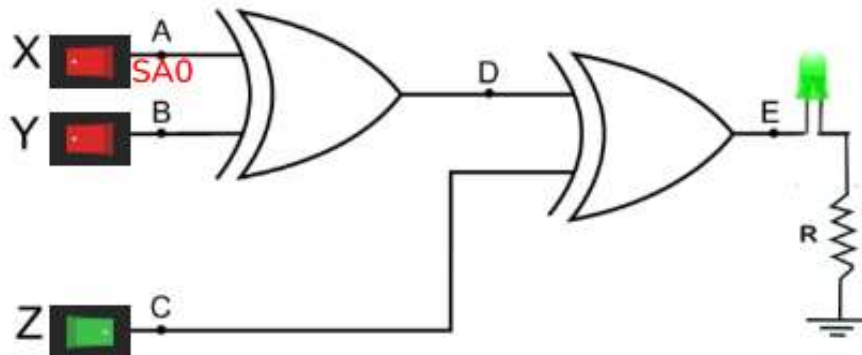
Inputs			outputs
W	X	Y	$Q = A \oplus B \oplus C$
0	0	0	0
0	0	1	1
0	1	0	1
0	1	1	0
1	0	0	1
1	0	1	0
1	1	0	0
1	1	1	1

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Simulation:

Paste your simulation result screen shot here:

Simulation 1



Simulation 2

Truth Table:

Input			Fault	Output (without Fault)	Output (with Fault)
X	Y	Z			
0	0	0	N/A	0	0
1	0	0	A SA0	1	0
1	0	0	A SA1	1	1

Conclusion:

In this virtual lab, we studied Stuck-at-0 (SA0) and Stuck-at-1 (SA1) faults in digital circuits. A Stuck-at-0 fault forces a line permanently to logic 0, while a Stuck-at-1 fault forces it to logic 1. By applying different test inputs, we observed how faulty outputs differ from normal outputs. The experiment showed that proper test vectors are required to detect these faults. Stuck-at fault modeling helps in identifying manufacturing defects. Overall, the lab improved our understanding of fault detection and circuit reliability.

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Post Lab Exercise

1. Which are the properties of a single stuck-at fault in digital circuits?

- (A) Only one line is faulty
- (B) The faulty line is permanently set to 0 or 1
- (C) The fault can be placed at an input or output of a gate
- (D) All of these ←**

[d]

2. When will the output always be 1 in 2-input XOR gate?

- (A) When both the inputs are same
- (B) When both the inputs are different ←**
- (C) All the above answers
- (D) None of these

3. A stuck-at fault may be introduced at any point within a digital circuit, including both the input and output nodes of logic gates?

- (A) True ←**
- (B) False

4. Which of the following options represents a real-world application of an XOR gate?

- (A) Radio
- (B) Switch Board
- (C) Game Show Buzzer ←**
- (D) Transformer

5. How many inputs are required to define a stuck-at fault concept?

- (A) Minimum 1, Maximum 3
- (B) Minimum 1, Maximum 5

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(C) Minimum 2, Maximum n

(D) Minimum 1, Maximum n ←

6. Stuck-at-0 or Stuck-at-1 is type of resistance in a circuit?

(A) True

(B) False ←

7. Which logic gate is described by the statement: 'Must have one or the other, but not both'?

(A) OR

(B) AND

(C) XOR ←

(D) XNOR

8. For an XOR gate with three or more input variables, the output will be 1.

(A) If an even number of variables must be equal to 1

(B) If an odd number of variables must be equal to 1 ←

(C) Both the above options

(D) We cannot determine with the given data

9. Which of the following Boolean expressions correctly represents the logic function of a 2-input XOR (Exclusive OR) gate?

(A) $A'B + AB'$ ←

(B) 3 bits

(C) $A + A.B$

(D) $A' + B'$

10. What are the possible conditions for a wire in a digital circuit?

(A) Stuck-at-0-fault

(B) Stuck-at-1-fault

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(C) No fault

(D) All of these←